

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
AI and application Development and jira

Practical – 7-B & C

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Aim:-Q.1Area (sq.ft.) Price (Lakhs)

600	32
800	40
1000	49
1200	58
1400	68
1600	79

1. Independent Variable (X)

Years of Experience

2. Dependent Variable (Y)

Salary

3. Why?

- Salary depends mainly on experience.
- Experience increases over time and generally leads to higher salary.
- Hence,
 - X = Years of Experience
 - Y = Salary

Code:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression
# Dataset
data = {
    "Area": [600, 800, 1000, 1200, 1400, 1600],
    "Price": [32, 40, 49, 58, 68, 79]
}
df = pd.DataFrame(data)
print(df)
```

```

# Independent and Dependent variables
X = df[["Area"]]
y = df["Price"]

# Train model
model = LinearRegression()
model.fit(X, y)
# Slope and Intercept
m = model.coef_[0]
c = model.intercept_

print("\nSlope =", m)
print("Intercept =", c)

print("\nRegression Equation:")
print(f"Price = {m:.4f} × Area + {c:.4f}")

# Predictions
price1500 = model.predict([[1500]])
price2500 = model.predict([[2500]])

print("\nPredicted Price for 1500 sq.ft =", round(price1500[0],2), "Lakhs")
print("Predicted Price for 2500 sq.ft =", round(price2500[0],2), "Lakhs")

```

Output:-

```

Area  Price
0    600    32
1    800    40
2   1000    49
3   1200    58
4   1400    68
5   1600    79

Slope = 0.04685714285714286
Intercept = 2.790476190476191

Regression Equation:
Price = 0.0469 × Area + 2.7905

Predicted Price for 1500 sq.ft = 73.08 Lakhs
Predicted Price for 2500 sq.ft = 119.93 Lakhs

```

Q.2 A company has collected the following employee data:

- Employee ID

- Employee Age
- Years of Experience
- Salary

You need to develop a Linear Regression model to predict an employee's salary.

Tasks

1. Identify the independent variable (X) and the dependent variable (Y).
2. Explain why you selected these variables.
3. Create a Pandas DataFrame using the given data.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Predict the salary for an employee having 7 years of experience.
7. Display:
 - o Slope (Coefficient)
 - o Intercept
 - o R2 Score
8. Plot the regression line.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Create DataFrame
data = {
    "Employee ID": [1, 2, 3, 4, 5, 6, 7, 8],
    "Age": [22, 24, 26, 28, 30, 32, 34, 36],
    "Experience": [1, 2, 3, 4, 5, 6, 8, 10],
    "Salary": [25000, 30000, 35000, 40000, 45000, 50000, 60000, 70000]
}

df = pd.DataFrame(data)
print(df)

# Independent and Dependent variables
X = df[["Experience"]]
Y = df["Salary"]

# Split dataset
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

```

# Train model
model = LinearRegression()
model.fit(X_train, Y_train)
# Predict salary for 7 years experience
salary = model.predict([[7]])
print("\nPredicted Salary for 7 Years Experience =", salary[0])

# Testing prediction
Y_pred = model.predict(X_test)
# Display results
print("\nSlope (Coefficient) =", model.coef_[0])
print("Intercept =", model.intercept_)
print("R2 Score =", r2_score(Y_test, Y_pred))

# Plot Regression Line
plt.scatter(X, Y)
plt.plot(X, model.predict(X))
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Experience vs Salary")
print("Aashu")
plt.show()

```

Output:-

```

Employee ID  Age  Experience  Salary
0           1   22           1   25000
1           2   24           2   30000
2           3   26           3   35000
3           4   28           4   40000
4           5   30           5   45000
5           6   32           6   50000
6           7   34           8   60000
7           8   36          10   70000

Predicted Salary for 7 Years Experience = 55000.0

Slope (Coefficient) = 5000.0
Intercept = 20000.0
R2 Score = 1.0
Aashu

```

